

ECOMMERCE

Data Analysis Project Documentation

Dataset: E-Commerce Public Dataset by Olist (Kaggle)

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Stack: SQL Server - Power BI - Power Query - DAX

1.Executive Summary

This project analyzes more than 100,000 orders from a Brazilian e-commerce marketplace (Olist) to extract actionable insights about sales, customers, sellers, logistics, and satisfaction. The analysis covers 2016–2018 and integrates data from 8 different sources, processed from SQL Server through interactive Power BI dashboards.

Key Business Metrics

\$8.70M Total Sales +19.0% YoY	53K Unique Customers +19.8% YoY	54K Order Volume +18.9% YoY	\$161.07 Average Order Value +0.1% YoY
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Key Findings

Geographic concentration in the Southeast

São Paulo leads with 40,300 customers (~41% of the market). The states of SP, RJ, and MG account for more than 60% of customers. Opportunity: expand into the North and Northeast, regions with lower penetration and substantial population potential.

The November effect — Black Friday

A sharp behavioral shift occurs: Fridays in November become the highest-sales day and the strongest day for acquiring new customers. This requires differentiated logistics and inventory pre-positioning during the two weeks before the last Friday of November.

Dissatisfaction in 'In Process' orders

The largest volume of negative reviews occurs for orders in the 'In Process' status, before they reach the carrier. The source document identifies this as an internal operational issue rather than an external logistics issue, with a recommended maximum SLA of 48 hours for this status.

Healthy delivery SLA: 90.8%

90.8% of the 99,000 orders were delivered within the estimated time. Improvement focus: Northeast states AL (21.4%), MA (17.4%), and SE (15.2%).

2. Project Context

Data Source

Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle). Información real anonimizada de un marketplace brasileno, cubriendo el periodo septiembre 2016 a octubre 2018.

Tabla SQL	Descripción
orders	Estado y fechas de cada orden (99,441 registros)
customers	Información demográfica y geográfica de clientes
order_items	Detalle de productos por orden (precio, flete)
order_payments	Método y valor de pago por orden
order_reviews	Puntuación y comentarios de clientes
products	Catálogo de productos y categorías
sellers	Información y ubicación de vendedores
geolocation	Coordenadas de códigos postales de Brasil

Project Objectives

Demonstrate the complete flow of a data project: from ingestion and modeling in SQL Server to executive visualization in Power BI, answering real business questions across 6 areas.

Area	Questions Answered
1. Customers and market	Geographic concentration and customer-base growth
2. Sales and products	Categories, average order value, and seasonality patterns
3. Sellers	Performance ranking, market concentration, and logistics
4. Logistics and deliveries	Delivery times, SLA, and critical delay zones
5. Payments and billing	Payment methods, ticket by type, and payment modalities

6. Customer satisfaction	Reviews, correlation with timing, and service quality
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3. Technical Architecture

Tecnology Stack

Tool	Role in the Project
SQL Server (T-SQL)	Database: ingestion, relational modeling, 6 analytical views, 18 business queries
Power BI Desktop	Dimensional modeling, relationships, DAX measures, and dashboard design
Power Query (M)	Column cleaning and normalization; dynamic Calendar table
DAX	Measures: YoY, YTD, average order value, market concentration, USERRELATIONSHIP
Excel	Rapid exploratory validation and review of master tables

Project Flow — 8 Phases

Phase	Activity	Tool
1	CSV loading + referential integrity (PK/FK)	SQL Server + SSMS
2	Geolocation normalization + indexes	SQL Server
3	6 analytical views as an intermediate layer	T-SQL
4	18 business queries over the T-SQL views	T-SQL
5	ODBC connection + Power BI load using native SQL	Power BI + ODBC
6	Cleaning, normalization, and Calendar table	Power Query (M)
7	Hybrid dimensional modeling and relationships	Power BI Model
8	DAX measures and interactive visualizations	Power BI DAX

Power BI Data Model

El modelo evolucionó de una estrella clásica a una arquitectura híbrida centrada en `dim_ordenes_detalle` como hub central. Esta tabla concentra clientes, vendedores, fechas, status y métricas de reseñas, evitando duplicaciones y simplificando el modelo.

Table	Type and Description
fact_orders_items	FACT TABLE — Item-level granularity: price, freight_value, product_id, seller_id
dim_ordenes_detalle	CENTRAL HUB — Order dates, status, avg/min/max reviews by order_id
dim_cliente	DIMENSION — Geographic data: customer_id, city, state, ZIP + coordinates
dim_vendedores	DIMENSION — seller_id, city, state, seller ZIP
dim_productos	DIMENSION — product_id and product category
dim_ordenes_pago	PAYMENTS — payment_type, payment_value, payment_sequential by order
Calendar	DATE TABLE — Complete hierarchies: Year, Quarter, Month, Day, Week

SQL Analytical Views

View	Purpose
vw_orders_detail	Base granular view: joins order, customer, seller, product, payment, and review
vw_info_clientes	Customer profile: total purchases, average order value, freight, and reviews
vw_info_vendedores	Seller performance: orders, sales, customers served, average review
vw_info_producto	Product metrics: min/max/avg price, units sold, cumulative total
vw_info_pagos	Payment-method usage: count, frequency, and total by city/state
vw_info_zonas	Zone logistics: delivery time, freight, satisfaction, and volume

4. Business Analysis Findings

4.1 Customers and Market

The market is strongly concentrated in Southeast Brazil. The three leading states account for more than 60% of total customers.

State	Unique Customers	Estimated Share
SP — São Paulo	40,300	~41%
RJ — Rio de Janeiro	12,380	~13%
MG — Minas Gerais	~7,200	~7%
RS — Rio Grande do Sul	5,280	~5%
PR — Paraná	~4,800	~5%
Customer-base growth (2016–2018)		
2016–2017: explosive growth of 13,308%, a pattern typical of a marketplace in a scaling phase. 2017–2018: sustained growth of 20.7%, indicating market consolidation. Source recommendation: capitalize on this momentum with expansion campaigns in the North and Northeast.		

4.2 Sales and Products

Category	Units Sold	
bed_bath_table (Bed, Bath & Table)	11,115	
health_beauty (Health & Beauty)	9,670	
sports_leisure (Sports & Leisure)	8,641	
furniture_decor (Furniture & Decor)	8,334	
computers_accessories (Computers)	7,827	
watches_gifts (Watches & Gifts)	5,991 — highest average order value (\$201.14)	

Seasonality and the Black Friday Effect

Sales peaks: March to June (sustained high season). Weekly activity is highest Monday through Thursday for most of the year. November is the exception: Fridays become the most active day.

Black Friday not only drives sales but also customer acquisition. Action: pre-position inventory and expand logistics capacity two weeks before the last Friday of November.

4.3 Sellers

Metric	Value
Leading seller	1,841 orders (ID: 6560211a...)
2nd seller	1,754 orders
Top 10 seller concentration	7.28% of total sales
Interpretation	Competitive and healthy marketplace (concentration < 10%)
Best individual delivery time	5.00 days average
Overall average delivery	12.5 days

4.4 Logistics and Deliveries

Metric	Value
SLA achieved	89,944 of 99,000 orders (90.8%)
Minimum delivery time by state	8.70 days — State SP
Maximum delivery time by state	29.34 days — State RR
AL (Alagoas) — highest delay rate	21.41% of orders outside SLA
MA (Maranhão)	17.43% of orders outside SLA
SE (Sergipe)	15.22% of orders outside SLA

4.5 Payments and Billing

Payment Method	Average Order Value
Credit Card (most used method)	\$163.32
Boleto Bancário (2nd method)	\$142.57

Debit	\$142.57
Voucher	\$65.70
Single payment vs installments	96.94% single payment — 3.06% installments

4.6 Customer Satisfaction

Finding	Detail
Overall average score	4.09 / 5.00 for delivered orders
AM case — highest satisfaction	4.21 average despite 26.36 days average delivery time
RR case — lowest satisfaction	3.61 average with 29.34 days average delivery time
Largest source of dissatisfaction	'In Process' status — internal operational issue
Key conclusion	Attention during the waiting period can mitigate the impact of late deliveries

5. Conclusions and Recommendations

5.1 Strategic Conclusions

Conclusion 1 — Growth is real but geographically concentrated

The marketplace grew +20.7% YoY. However, 41% of customers are in a single state (SP). Recommendation: expansion campaigns toward the North and Northeast, regions with lower penetration and high population. Success metric: increase the share of Northern states from the current 5% to 12% within 18 months.

Conclusion 2 — The satisfaction issue is operational, not logistical

The worst reviews are associated with orders remaining in 'In Process' too long rather than with delivery delays. Recommendation: establish a maximum internal SLA of 48 hours for 'In Process'. Complementary action: proactively notify customers when an order exceeds that limit.

Expected impact: reduce the percentage of negative reviews for 'In Process' orders by 30–40%.

Conclusion 3 — Black Friday requires differentiated logistics planning

In November, behavior changes sharply: Fridays outperform all other days. Recommendation: pre-position inventory and expand logistics capacity during the two weeks before Black Friday. Additional action: activate acquisition campaigns specifically for November Fridays.

Conclusion 4 — Northeast states require logistics attention

AL, MA, and SE have delay rates of 15–21%, above the national average of 9.2%. Recommendation: evaluate partnerships with regional logistics operators. Alternative: establish intermediate distribution centers in the Northeast.

5.2 Project Improvement Roadmap

Technical Improvement	Expected Impact
Implement RLS (Row Level Security)	Segmented seller access in the dashboard
Power Automate alerts	Automatic notification if AL or MA SLA falls below 85%
Incorporate CAC and marketing data	Calculate real ROI by product category
Executive Summary page	View of 5 findings for a non-technical audience
Reclassify dim_orderes_pago as a fact table	Improved data-model integrity

6. About the Author

Tushar Rana

Data Analyst | SQL Server - Power BI - Power Query - DAX

Skill	Level and Description
SQL Server (T-SQL)	Relational modeling, views, composite indexes, stored procedures
Power BI + DAX	Dimensional modeling, USERRELATIONSHIP, YoY/YTD, dashboard UX/UI design
Power Query (M)	ETL, data cleaning, Calendar tables, advanced transformations
Excel	Pivot tables, Power Query, Power Pivot, interactive dashboards
Business analysis	Translating business questions into queries and executive visualizations
Applied analytics	Understanding consumer behavior and interpreting qualitative data

This project is part of a professional portfolio built during a career transition toward data analysis. It combines technical rigor with analytical thinking focused on real business impact.

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